Listing 6.6 Plant the Grass tests

```
Test.assert_equals(simulate_grass(
["xxxxxxx","xoooxox","xxxxoox","xoooxxx","xxxxxxx"],
 1, 1),
["xxxxxxx","x+++++x","xxxx++x","xoooxxx","xxxxxxx"])
    Test.assert_equals(simulate_grass(
["xxxxxxx","xoxoxox","xxoooox","xooxxxx",
"xoxooox","xoxoxox","xxxxxxx"],
 2, 3), ["xxxxxxx","xox+++x","xx++++x","x++xxxx",
"x+xooox","x+xooox","xxxxxxx"])
    Test.assert_equals(simulate_grass(
["xxxxxx","xoxoox","xxooox","xoooox","xoooox","xxxxxx"],
1, 1),
["xxxxxx","x+xoox","xxooox","xoooox","xoooox","xxxxxx"])
    Test.assert_equals(simulate_grass(
["xxxxx","xooox","xooox","xooox","xxxxx"],
1, 1),
["xxxxx","x+++x","x+++x","x+++x","xxxxx"])
    Test.assert_equals(simulate_grass(
["xxxxxx","xxxxox","xxooox","xoooxx","xooxxx",
"xooxxx","xxooox","xxxoxx","xxxxxx"],
4, 1),
["xxxxxx","xxxx+x","xx+++x","x+++xx","x++xxx",
"x++xxx","xx+++x","xxx+xx","xxxxxx"])
    Test.assert_equals(simulate_grass(
["xxxxxxxxxxx", "xoxooooooox", "xoxoxxxxxox",
"xoxoxoooxox", "xoxoxoxoxox", "xoxoxoxoxox",
"xoxoxxxoxox", "xoxooooxox", "xoxxxxxxxox",
"xooooooooox", "xxxxxxxxxxx"], 1, 1),
["xxxxxxxxxxx", "x+x++++++x", "x+x+xxxxx+x",
"x+x+x+++x+x", "x+x+x+x+x+x", "x+x+x+x+x+x",
"x+x+xxx+x+x", "x+x++++++x", "x+xxxxxxx+x",
"x++++++++x", "xxxxxxxxxxx"])
```

The tests will be run as part of a two-step verification to confirm that the solution works. We'll also use the tests and challenges as written, which will further test the AI.

Figure 6.8 shows the makeup of a straightforward behavior tree that will be used to solve various programming challenges. You'll notice that this ABT uses a different assistant for the actions and conditions. For the first step, the Python coding assistant (called the Hacker) generates a solution that is then reviewed by the coding challenge Judge (called the Judge), which produces a refined solution that is verified by a different Python coding assistant (called the Verifier).